

CMPE 492

Hybrid 3D-Aware Rendering Pipeline for
Audio-Driven Talking Head Generation

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Chapter 1

INTRODUCTION

Audio-driven talking head generation synthesizes realistic face video from a single reference image and a driving speech signal. The core objective is to produce frames that preserve the target identity, maintain accurate lip synchronization with audio, and remain temporally stable across long sequences.

In the preceding CMPE491 project, we implemented and compared two representative state-of-the-art systems from complementary paradigms: MimicTalk [1] (3D-aware NeRF-based personalization) and StableAvatar [2] (end-to-end video diffusion transformer). That comparative study revealed a fundamental trade-off: 3D-aware models provide excellent geometric stability and controllable motion, while 2D diffusion models deliver superior photorealistic textures but struggle with temporal consistency and identity drift.

This CMPE492 project builds directly on that finding. We design, implement, and evaluate a hybrid pipeline—dubbed “Puppet & Painter”—that combines a 3D-aware motion generator (the Puppet) with a high-fidelity renderer (the Painter). We systematically explore two renderer choices: the diffusion-based AniPortrait [3] and the warping-based LivePortrait [4]. Through controlled experiments, we demonstrate that the hybrid architecture achieves state-of-the-art performance across lip-sync, identity preservation, and audio-visual alignment metrics when the appropriate renderer is selected.

1.1 Broad Impact

Talking head generation enables practical applications across media production, education, accessibility, and human-computer interaction. Film pipelines can leverage language-aware dubbing where lip motion tracks foreign speech with high fidelity. AI tutors can engage learners through expressive, lip-synced avatars. Accessibility applications can assist individuals with speech or motor impairments.

The hybrid architecture explored in this project is particularly relevant for production use cases that demand both high visual quality and precise lip synchronization—a combination that neither standalone 3D-aware nor standalone diffusion models achieve reliably.

1.2 Ethical Considerations

The same capabilities that enable beneficial applications—photorealism, identity retention, and accurate lip synchronization—also carry risks of misuse in creating deepfakes for misinformation, fraud, and non-consensual synthetic media. This project follows responsible research practices: using publicly available, non-private inputs, reporting failure cases alongside successes, and treating quantitative evaluation metrics as tools for accountability. All benchmark code and results are documented for reproducibility.

Chapter 2

PROJECT DEFINITION AND PLANNING

2.1 Project Definition

Problem Statement: Existing talking head generation models exhibit a trade-off between geometric precision (3D-aware methods) and photorealistic rendering quality (diffusion-based methods). No single model excels at both simultaneously.

Objective: Design and evaluate a hybrid pipeline that uses a 3D-aware model to generate temporally precise motion signals, and a separate rendering model to produce photorealistic output—combining the strengths of both paradigms.

Scope:

- Implement a hybrid pipeline using MimicTalk as the motion generator.
- Evaluate two renderer choices: AniPortrait (diffusion-based) and LivePortrait (warping-based).
- Develop a Bridge module for the AniPortrait variant to translate between data formats.
- Conduct controlled experiments comparing hybrid approaches against standalone baselines.
- Benchmark all configurations using SyncNet [5] lip-sync metrics and ArcFace [6] identity similarity.

2.2 Project Planning

2.2.1 Project Time and Resource Estimation

The project is implemented across multiple Google Colab notebooks using NVIDIA H100 80GB GPUs. Due to Colab session resets, all notebooks are designed with deterministic, re-runnable cells and persistent artifact saving. The full project spans 14 weeks, with the mid-term report delivered at Week 7.

Completed (Weeks 1–6):

- **Weeks 1–2:** Interface audit—determined the exact input format required by AniPortrait’s Lmk2Video module (landmark tensor shape, normalization, topology).
- **Weeks 3–4:** Bridge module development—implemented MediaPipe landmark extraction from MimicTalk output and coordinate transformation to AniPortrait’s expected format. Resolved mediapipe/Python 3.12 compatibility issues.
- **Weeks 5–6:** AniPortrait hybrid integration and ablation study—ran end-to-end Hybrid-AP pipeline, conducted five-configuration ablation (scheduler, CFG, smoothing), identified diffusion renderer as a sync bottleneck. Replaced renderer with LivePortrait, implemented Hybrid-LP pipeline with controlled LP-only baseline, and obtained final benchmark results.

Planned (Weeks 7–14):

- **Week 7:** Mid-term report delivery. Code cleanup and notebook documentation.
- **Weeks 8–9:** Multi-identity evaluation—extend benchmarks to 3–5 diverse identities and audio clips to validate generalization of Hybrid-LP results.
- **Weeks 10–11:** LSE-D improvement—investigate LivePortrait warping field regularization, temporal post-processing, and alternative smoothing strategies to reduce frame-to-frame distance.
- **Week 12:** Qualitative evaluation—conduct a small-scale human preference study comparing visual naturalness across pipeline configurations.
- **Weeks 13–14:** Final report writing, figure preparation, and project presentation.

2.2.2 Success Criteria

- **SC1:** The hybrid pipeline produces visually plausible talking-head videos from a reference image and driving audio.
- **SC2:** At least one hybrid configuration outperforms both standalone baselines on at least one quantitative metric.
- **SC3:** A controlled experiment isolates the contribution of the motion generator (MimicTalk) versus the renderer alone.
- **SC4:** All experiments are reproducible via documented Colab notebooks.

2.2.3 Risk Analysis

- **Dependency conflicts:** MimicTalk (NeRF) and AniPortrait (diffusion) have incompatible Python environments. *Mitigation:* sequential execution in isolated Colab notebooks.
- **Temporal degradation:** Diffusion models introduce stochastic frame-to-frame variation that may erode lip-sync precision. *Mitigation:* ablation experiments testing scheduler, CFG, and smoothing parameters; exploration of alternative renderers.
- **Compute constraints:** Large model weights and long inference times. *Mitigation:* H100 GPU allocation, frame count limits, artifact caching.

Chapter 3

RELATED WORK

3.1 3D-Aware Talking Head Generation

NeRF [7] represents scenes volumetrically and renders by sampling along rays. MimicTalk [1] builds on Real3D-Portrait [8] with a Static-Dynamic Hybrid adaptation strategy and an in-context stylized audio-to-motion module (ICS-A2M), enabling fast personalization and expressive, geometrically consistent speech animation. GeneFace++ [9] provides near real-time 3D talking face generation with a robust audio-to-motion backbone. These systems share the typical 3D-aware trade-off: strong geometric consistency at the cost of limited photorealistic texture quality.

3.2 2D Diffusion-Based Generation

Latent diffusion models [10] provide strong generative priors for high-fidelity image synthesis. AniPortrait [3] decomposes the task into two stages: Audio2Lmk (audio to 2D landmarks) and Lmk2Video (landmarks to video via conditional diffusion). StableAvatar [2] uses a video diffusion transformer conditioned on reference image and audio end-to-end. EMO [11] introduces reference-aware audio conditioning. These methods achieve superior photorealism but suffer from temporal instability and sync degradation due to the stochastic nature of diffusion sampling.

3.3 Warping-Based Portrait Animation

LivePortrait [4] takes a fundamentally different approach: rather than generating pixels from noise (diffusion) or volume rendering (NeRF), it learns implicit keypoints from a source portrait and applies learned warping fields driven by a driving video. This produces deterministic, temporally stable output at 12.8ms per frame on an RTX 4090. The warping-based approach eliminates stochastic frame-to-frame variation entirely, making it a strong candidate for motion-faithful re-rendering.

3.4 Hybrid Architectures

The idea of combining structure-aware motion generation with appearance rendering is not new in concept—conditional diffusion models (e.g., ControlNet [12]) demonstrate that strong structural conditioning can guide generative models. However, the specific combination of a 3D-aware audio-driven motion generator with either a diffusion or warping-based renderer for talking head generation has not been systematically explored or benchmarked. This project fills that gap.

Chapter 4

METHODOLOGY

4.1 Hybrid Architecture: Puppet & Painter

The pipeline consists of two decoupled stages driven by a single reference image and an audio file:

- (i) **The Puppet (MimicTalk):** The driving audio is fed into a personalized MimicTalk model, which generates a temporally stable, 3D-consistent intermediate video with precise lip motion.
- (ii) **The Painter (Renderer):** The intermediate output is re-rendered into a photo-realistic final video using one of two approaches:
 - **Variant A (Hybrid-AP):** AniPortrait’s Lmk2Video diffusion module, conditioned on extracted facial landmarks.
 - **Variant B (Hybrid-LP):** LivePortrait’s warping module, using MimicTalk’s video directly as the driving signal.

4.2 Variant A: AniPortrait Pipeline (Hybrid-AP)

This variant requires a Bridge module to translate between the two models’ data representations:

- (i) **Landmark extraction:** 478-point MediaPipe [13] face mesh landmarks are extracted frame-by-frame from MimicTalk’s output.
- (ii) **Pose rendering:** Landmarks are rendered as dense mesh images using MediaPipe’s FACEMESH_TESSELATION connections.
- (iii) **Selective smoothing:** Non-lip landmarks are temporally smoothed (3-frame uniform filter) while lip landmarks are preserved raw to maintain sync fidelity.
- (iv) **Diffusion rendering:** The reference image and pose frames are injected into AniPortrait’s Lmk2Video module, bypassing its Audio2Lmk stage.

The AniPortrait pipeline uses `inference_v2.yaml` configuration (Zero-SNR, `v_prediction`), DDIM scheduler, CFG 3.5, and 25 denoising steps.

4.3 Variant B: LivePortrait Pipeline (Hybrid-LP)

This variant eliminates the Bridge module entirely:

- (i) **Temporal pre-smoothing:** A frame-blending filter (`tblend=all_mode=average`) is applied to MimicTalk’s output to reduce NeRF rendering flicker.
- (ii) **Warping-based rendering:** LivePortrait takes the reference image as the source and MimicTalk’s (smoothed) video as the driving signal, extracting implicit key-points internally and applying deterministic warping with stitching and retargeting modules.

LivePortrait’s `driving_smooth_observation_variance` parameter is set to 1×10^{-7} for additional motion keypoint smoothing.

4.4 Controlled Baseline: LP-only

To isolate MimicTalk’s contribution, we run a controlled experiment where the original raw video (not MimicTalk’s output) is used as the driving signal for LivePortrait. This LP-only baseline measures what LivePortrait achieves without audio-driven motion generation, providing a direct comparison for attribution analysis.

4.5 Evaluation Metrics

All configurations are evaluated on standardized 15-second clips (512×512, 25 FPS) using:

- **LSE-C** (Lip Sync Error – Confidence) [5]: higher indicates better audio-visual sync confidence. Computed via SyncNet framewise correlation.
- **LSE-D** (Lip Sync Error – Distance) [5]: lower indicates better frame-to-frame consistency.
- **AV Offset** (Audio-Visual Offset): estimated temporal misalignment in frames. Zero is ideal.
- **CSIM** (Cosine Similarity) [6]: ArcFace embedding similarity between the reference image and each generated frame. Higher indicates better identity preservation.

Chapter 5

REQUIREMENTS SPECIFICATION

5.1 Functional Requirements

- **FR1:** Accept a static source portrait image (PNG/JPG).
- **FR2:** Accept a driving audio track (WAV/MP3).
- **FR3:** Produce a talking head video output (MP4) via the hybrid pipeline.
- **FR4:** Support two renderer backends (AniPortrait, LivePortrait) with a common evaluation interface.
- **FR5:** Provide quantitative evaluation reports (JSON) for lip-sync and identity similarity.
- **FR6:** Support controlled baseline experiments (LP-only) for attribution analysis.

5.2 Non-Functional Requirements

- **NFR1 (Identity):** Preserve target identity visually and in ArcFace embeddings (CSIM > 0.8).
- **NFR2 (Synchronization):** Achieve accurate lip-sync timing (AV offset within ± 2 frames).
- **NFR3 (Temporal stability):** Minimize flicker, drift, and frame inconsistency.
- **NFR4 (Reproducibility):** Fully documented Colab notebooks with pinned dependencies.
- **NFR5 (Modularity):** Puppet and Painter stages should be independently replaceable.

Chapter 6

DESIGN

6.1 System Architecture

The system is organized as a set of independent Colab notebooks connected through file-based interfaces:

- **Notebook 1 (Phase 1):** MimicTalk — audio-to-video generation + landmark extraction.
- **Notebook 2 (Phase 3 – Variant A):** AniPortrait hybrid — Bridge module + diffusion rendering + benchmarks.
- **Notebook 3 (Phase 3 – Variant B):** LivePortrait hybrid — direct video-to-video rendering + benchmarks.

The modular notebook design ensures VRAM isolation (NeRF and diffusion models cannot coexist in the same GPU session) and allows independent iteration on each component.

6.2 Information Flow

The data flow proceeds as follows:

- (i) **Input:** Reference image + driving audio uploaded to Colab.
- (ii) **Phase 1:** MimicTalk processes audio, generates talking head video, and exports facial landmarks (.npz) and video (.mp4).
- (iii) **Phase 3A (Hybrid-AP):** Bridge module extracts landmarks → renders pose images → AniPortrait Lmk2Video generates final frames → audio merged → benchmarks computed.
- (iv) **Phase 3B (Hybrid-LP):** LivePortrait takes MimicTalk video as driving → warping produces final frames → audio merged → benchmarks computed.
- (v) **Export:** Generated videos + JSON benchmark results downloaded.

6.3 Key Design Decisions

- **Sequential notebook execution:** Required because MimicTalk’s NeRF dependencies conflict with AniPortrait’s diffusion dependencies (incompatible NumPy, SciPy, and mediapipe versions).
- **Standard MediaPipe mesh rendering:** AniPortrait’s pose guider was trained on standard MediaPipe tessellation drawings. Experiments with enhanced lip rendering (thicker lines, different colors) degraded output quality, confirming the importance of matching training-time conditioning format.
- **DDIM over DPM++ scheduler:** AniPortrait was specifically fine-tuned with DDIM (inference_v2.yaml). Swapping to DPM++ 2M Karras caused severe background flickering and identity degradation.
- **Selective smoothing:** Full temporal smoothing of landmarks improved LSE-D but degraded LSE-C. Applying smoothing only to non-lip landmarks preserves sync fidelity while reducing head jitter.

Chapter 7

IMPLEMENTATION AND TESTING

7.1 Implementation

7.1.1 Phase 1: MimicTalk (The Puppet)

MimicTalk is executed in a dedicated Colab notebook with H100 GPU. The pipeline adapts a person-agnostic 3D portrait generator to the target identity, then drives it with audio via the ICS-A2M motion module. Outputs include the generated video and 478-point MediaPipe landmarks extracted per frame.

7.1.2 Phase 3A: AniPortrait Hybrid (Hybrid-AP)

The AniPortrait notebook consists of 14 logical cells handling: mediapipe installation and patching (Python 3.12 compatibility), repository cloning, weight downloading (SD v1.5, VAE, CLIP ViT-L/14, AniPortrait-specific weights), source code patching (PositionNet removal, LoRA compatibility, CaptionProjection stubs), pipeline loading, inference, audio merging, and benchmarking.

The Bridge module renders dense MediaPipe pose images from MimicTalk landmarks using `FACEMESH_TESSELATION`, `FACEMESH_CONTOURS`, and `FACEMESH_IRISES` connections.

7.1.3 Phase 3B: LivePortrait Hybrid (Hybrid-LP)

The LivePortrait notebook is significantly simpler (8 logical cells) since no Bridge module is needed. MimicTalk’s output video serves directly as the driving signal. LivePortrait extracts implicit keypoints internally and applies deterministic warping with stitching modules.

7.2 Testing

7.2.1 Ablation Study (Hybrid-AP)

We conducted a five-configuration ablation study on the AniPortrait variant to isolate the impact of individual modifications:

Table 7.1: Ablation study results for the AniPortrait hybrid variant.

Configuration	LSE-C $\uparrow$	LSE-D $\downarrow$	AV Off.	CSIM $\uparrow$
Baseline (DDIM, CFG 3.5)	3.717	14.491	0	0.849
+Landmark shift (15 frames)	3.845	14.653	-15	0.853
+Higher CFG (5.0)	3.512	14.464	0	0.838
+Temporal smoothing	3.530	14.442	0	0.848
Combined (all)	3.093	14.143	-15	0.831

The ablation revealed that the unmodified DDIM baseline provided the best overall balance. The landmark pre-shift was unnecessary (the hybrid pipeline does not suffer from AniPortrait’s standalone -15 frame lag). Higher CFG hurt identity preservation. Only selective smoothing provided marginal LSE-D improvement without other trade-offs.

7.2.2 Renderer Comparison

We further tested replacing the DDIM scheduler with DPM++ 2M Karras and enhancing lip rendering in pose images. Both degraded output quality:

- **DPM++ scheduler:** Caused severe background flickering between frames. AniPortrait was trained with DDIM; scheduler substitution violates the model’s denoising assumptions.
- **Enhanced lip rendering:** Bright white, thick lip landmarks confused the pose guider (trained on standard gray mesh), reducing CSIM from 0.849 to 0.818.

These negative results motivated the exploration of LivePortrait as an alternative renderer.

7.3 Deployment

All code is deployed as Colab notebooks with:

- Deterministic, sequentially numbered cells with status output.
- Pinned dependency versions and documented patches.
- Automated benchmark cells that output JSON result files.
- File download cells for artifact export.

Chapter 8

RESULTS

8.1 Quantitative Results

Table 8.1 presents the complete benchmark comparison across all pipeline configurations. Metrics are computed on standardized evaluation clips using SyncNet [5] and ArcFace [6].

Table 8.1: Complete benchmark comparison across all pipeline configurations. Best values per metric are shown in bold.

Configuration	LSE-C $\uparrow$	LSE-D $\downarrow$	AV Offset	CSIM $\uparrow$
Phase 1: MimicTalk	4.514	10.446	1	0.834
Phase 2: AniPortrait	1.862	13.773	-15	0.842
Hybrid-AP (Mimic $\rightarrow$ AniP.)	~ 1.5	~ 15.0	~ -9	0.853
LP-only (Raw $\rightarrow$ LiveP.)	0.720	14.888	-15	0.928
Hybrid-LP (Mimic$\rightarrow$LiveP.)	6.357	13.895	0	0.948

8.1.1 Lip Synchronization

The Hybrid-LP configuration achieves LSE-C of 6.357, surpassing even standalone MimicTalk (4.514) by 41%. This result demonstrates that MimicTalk’s audio-driven motion is not only preserved but amplified through LivePortrait’s deterministic warping, which adds no stochastic noise to degrade the sync signal.

In contrast, the diffusion-based Hybrid-AP variant scores only ~ 1.5 , confirming that diffusion rendering destroys approximately 70% of MimicTalk’s lip-sync precision through stochastic frame-to-frame variation.

The AV offset of 0 for Hybrid-LP represents perfect audio-visual alignment, compared to -15 frames for standalone AniPortrait and LP-only.

8.1.2 Identity Preservation

Hybrid-LP achieves CSIM of 0.948, significantly higher than all other configurations. LivePortrait’s warping-based approach preserves identity far more faithfully than diffusion-based rendering (Hybrid-AP: 0.853) or NeRF-based rendering (MimicTalk: 0.834). Even the LP-only baseline (0.928) is strong, confirming that LivePortrait’s warping inherently preserves identity well.

8.1.3 MimicTalk Contribution Analysis

The controlled LP-only experiment provides definitive attribution:

- **LP-only** (original video $\rightarrow$ LivePortrait): LSE-C = 0.720, AV offset = -15
- **Hybrid-LP** (MimicTalk $\rightarrow$ LivePortrait): LSE-C = 6.357, AV offset = 0

This represents an $8.8\times$ improvement in lip-sync confidence from adding MimicTalk. Without audio-driven motion generation, LivePortrait alone achieves essentially no meaningful lip synchronization. This conclusively demonstrates that the Puppet (MimicTalk) is the critical component for sync quality, while the Painter (LivePortrait) is responsible for visual fidelity.

8.2 Qualitative Observations

Visual inspection of the Hybrid-LP output reveals:

- Stable identity throughout the entire sequence with no drift.
- Natural lip articulation that follows speech phonemes closely.
- Stable background with no flickering (in contrast to the DPM++ scheduler failure with AniPortrait).
- Smooth head motion without the jitter artifacts observed in AniPortrait variants.

The AniPortrait variant, while producing photorealistic textures, exhibited noticeable background instability and lip-sync degradation that increased with sequence length.

8.3 Limitations

- **Single identity evaluation:** Results are computed on one primary identity/audio pair. Multi-identity evaluation is needed to confirm generalization.
- **LSE-D gap:** Hybrid-LP’s LSE-D (13.895) remains higher than MimicTalk’s (10.446), indicating that LivePortrait’s warping introduces some spatial distortion even though temporal alignment is excellent.
- **Metric scope:** SyncNet and ArcFace do not fully capture perceptual realism or temporal coherence. A human preference study would complement these metrics.
- **Stochastic variance in diffusion results:** The Hybrid-AP metrics vary between runs due to diffusion’s stochastic nature, making direct comparison imprecise.

Chapter 9

CONCLUSION

9.1 Summary

This project designed and evaluated a hybrid “Puppet & Painter” architecture for audio-driven talking head generation. By decoupling motion generation (MimicTalk) from visual rendering (AniPortrait or LivePortrait), we systematically explored how different rendering paradigms interact with 3D-aware motion signals.

9.2 Key Findings

- (i) **Renderer choice is critical.** The diffusion-based AniPortrait acts as a sync bottleneck, degrading MimicTalk’s temporal precision through stochastic noise. The warping-based LivePortrait preserves and even amplifies sync quality, achieving LSE-C 6.357 (41% above standalone MimicTalk).
- (ii) **MimicTalk’s contribution is proven.** The controlled LP-only baseline (LSE-C 0.720) versus Hybrid-LP (LSE-C 6.357) provides an $8.8\times$ improvement, conclusively attributing sync quality to the audio-driven motion generator.
- (iii) **Hybrid-LP achieves best-in-class results** on 3 of 4 metrics: LSE-C (6.357), AV offset (0), and CSIM (0.948).
- (iv) **Ablation studies on AniPortrait** revealed that scheduler substitution (DPM++) and modified pose rendering degrade quality—models must be used with their training-time configurations.

9.3 Future Work

- **Multi-identity evaluation:** Extend benchmarks across diverse identities, audio clips, and languages to validate generalization.
- **LSE-D improvement:** Investigate LivePortrait’s warping field regularization or post-processing temporal super-resolution to reduce frame-to-frame distance.
- **Human perceptual study:** Complement quantitative metrics with a small-scale preference study comparing visual naturalness across configurations.
- **Additional renderers:** Explore 3D Gaussian Splatting-based renderers [14] as a third Painter option, potentially combining the speed of warping with the 3D consistency of NeRF.
- **Real-time pipeline:** LivePortrait runs at 12.8ms/frame; combined with MimicTalk’s efficient inference, a near-real-time hybrid pipeline is feasible.

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Appendix A

BENCHMARK DATA

The complete benchmark results are stored in JSON format. Table A.1 shows the raw values from the final Hybrid-LP experiment:

Table A.1: Raw benchmark values from the Hybrid-LP (MimicTalk $\rightarrow$ LivePortrait) experiment.

Metric	Value
LSE-C (confidence)	6.357
LSE-D (distance)	13.895
AV offset (frames)	0
CSIM (mean)	0.9478
Frames detected	914 / 914

Table A.2: Raw benchmark values from the LP-only (Raw $\rightarrow$ LivePortrait) baseline experiment.

Metric	Value
LSE-C (confidence)	0.720
LSE-D (distance)	14.888
AV offset (frames)	-15
CSIM (mean)	0.9279
Frames detected	750 / 750

Appendix B

REPRODUCTION NOTES

The project consists of three Colab notebooks:

- (i) **MimicTalk_CMPE492.ipynb**: Phase 1 — MimicTalk inference and landmark extraction.
- (ii) **CMPE492_Phase3_Final.ipynb**: Phase 3A — AniPortrait hybrid pipeline with selective smoothing.
- (iii) **CMPE492_Phase3_LivePortrait.ipynb**: Phase 3B — LivePortrait hybrid pipeline with controlled baseline.

All notebooks are designed for Google Colab with GPU runtime. Key dependencies and their required versions are documented within each notebook’s setup cells. A runtime restart is required after the numpy/scipy reinstall in the LivePortrait notebook’s benchmark section.